

Math727 – Numerical Analysis II

Midterm Exam

1. Let us consider the two-point BVP

$$-u''(x) = f(x), \quad x \in (0, 1),$$

$$u(0) = u(1) = 0.$$

- (a) If $f(x) = 1/x$, use the Green function theory to prove that $u(x) = -x \log x$.
- (b) Let $\mathbf{A}$ be the coefficient matrix of the system obtained by discretizing the above equation using the 2nd order central difference. Show that if $\mathbf{Ax} \geq \mathbf{0}$ for some vector $\mathbf{x}$, then $\mathbf{x} \geq \mathbf{0}$.
2. Let $\mathbf{A} \in \mathbb{R}^{N \times N}$ be a symmetric matrix with eigenvalues $\lambda_1 = \lambda_2 = \dots = \lambda_r$, $|\lambda_r| > |\lambda_{r+1}| \geq \dots \geq |\lambda_N|$. With the vector sequence $\mathbf{z}^{(k+1)} = \mathbf{Az}^{(k)}$, $k = 0, 1, \dots$, one forms the sequence of Rayleigh quotients:

$$\alpha_k = \frac{(\mathbf{z}^{(k)})^T \mathbf{z}^{(k+1)}}{\|\mathbf{z}^{(k)}\|_2^2}, \quad k = 0, 1, \dots,$$

where the starting vector $\mathbf{z}^{(0)}$ has a nonzero contribution in the eigenspace of $\mathbf{A}$ corresponding to the eigenvalue λ_1 . Show that for an eigenvector $\mathbf{x}$ corresponding to the eigenvalue λ_1 it holds that

$$\text{sgn}(\alpha_k)^k \frac{\mathbf{z}^{(k)}}{\|\mathbf{z}^{(k)}\|_2} = \mathbf{x} + O\left(\left|\frac{\lambda_{r+1}}{\lambda_1}\right|^k\right)$$

for $k \rightarrow +\infty$.

3. For a nonsingular matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$ and a vector $\mathbf{b} \in \mathbb{R}^N$, show that the following statements are equivalent for $n \geq 1$:
- (a) The vectors $\mathbf{b}, \mathbf{Ab}, \dots, \mathbf{A}^n \mathbf{b}$ are linearly dependent;
 - (b) $\mathcal{K}_n(\mathbf{A}, \mathbf{b}) = \mathcal{K}_{n+1}(\mathbf{A}, \mathbf{b})$.
 - (c) $\mathbf{A}(\mathcal{K}_n(\mathbf{A}, \mathbf{b})) \subset \mathcal{K}_n(\mathbf{A}, \mathbf{b})$.
 - (d) For $\mathbf{x}_* = \mathbf{A}^{-1} \mathbf{b}$ there holds $\mathbf{x}_* \in \mathcal{K}_n(\mathbf{A}, \mathbf{b})$.
4. Let $\mathbf{A} \in \mathbb{R}^{N \times N}$ be a SPD matrix and $\mathbf{b} \in \mathbb{R}^N$ be a vector. Consider the CG method for solution of the linear system $\mathbf{Ax} = \mathbf{b}$.
- (a) Show that if there are exactly $k \leq N$ distinct eigenvalues of $\mathbf{A}$, then the CG method will terminate in at most k iterations.
 - (b) Show that

$$\frac{\|\mathbf{r}_k\|_2}{\|\mathbf{b}\|_2} \leq \sqrt{\kappa_2(\mathbf{A})} \frac{\|\mathbf{x}^* - \mathbf{x}_k\|_A}{\|\mathbf{x}^*\|_A}$$

5. Let

$$\mathbf{A} = \begin{pmatrix} 0 & & & 1 \\ 1 & 0 & & \\ & \ddots & \ddots & \\ & & 1 & 0 \end{pmatrix} \in \mathbb{R}^{N \times N}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \mathbf{x}^* = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}.$$

Clearly $\mathbf{x}^*$ is the exact solution of the linear system $\mathbf{Ax} = \mathbf{b}$. Show how GMRES will converge on the linear system $\mathbf{Ax} = \mathbf{b}$.